Credit Markets and the Macroeconomy: a Vision and My Research Agenda

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I build quantitative models of credit markets and the banks that intermediate them to understand how they shape macroeconomic outcomes. Two questions run through my work: how do household and bank balance sheets alter the transmission of monetary and fiscal policy, and who bears the costs when regulators tighten capital requirements? My research lies at the intersection of macroeconomics, monetary and fiscal policy, and banking.

This essay organises my projects around the policy questions, gaps in the literature, and ideas that connected one to the next.

How did I start?

The same interest-rate cut produces very different effects depending on how much debt households carry. Household-debt-to-GDP roughly doubled across OECD economies between 1995 and 2008, yet the workhorse New Keynesian models used for policy analysis had no role for the household balance sheet.[1] That disconnect—between what practitioners saw and what the models could say—motivated my first projects and still drives my agenda. I wanted to know whether balance sheets matter quantitatively, not just in principle, and if so, through which channels.

[1] Central bankers at the RBA, Riksbank, and Norges Bank worried openly about leverage distorting policy transmission. The models they relied on could not speak to the concern.

Does household debt alter the transmission of monetary and fiscal policy?

Papers: Zurita (2026a, Macroeconomic Dynamics); Zurita (2024, Macroeconomic Dynamics).

I began by asking the most direct version of the question. Standard New Keynesian models predict that a monetary-policy easing stimulates output through the interest-rate channel, with the size of the effect pinned down by a handful of deep parameters. Household debt does not appear among them. Yet practitioners—at the RBA, the Riksbank, Norges Bank—routinely worry that high household leverage amplifies or distorts policy transmission. Is this concern justified?

In Zurita (2026a), I investigate the question using a smooth-transition vector autoregression (STVAR) estimated on data from Australia, Sweden, Norway, and the G7 economies. The smooth-transition framework allows the impulse responses themselves to depend on the state of household debt, rather than imposing a single linear response.[2] Two findings stand out. First, the *short-run* effects of a monetary easing are generally stronger during periods of high household debt—on impact, the stimulus is roughly

[2] A key advantage of the STVAR is that it nests the linear VAR as a special case. If debt does not matter, the data will say so.

0.06 percentage points of GDP larger in Norway and the United States when debt is elevated. The mechanism is intuitive: indebted households are more sensitive to changes in interest payments, so a rate cut releases more disposable income. Second, and more subtly, high household debt *diminishes the persistence* of the stimulus over the medium term (4–8 quarters), suggesting that the initial boost comes partly at the expense of future spending as debt-service obligations reassert themselves.

The fiscal-policy counterpart is in Zurita (2024). The same STVAR methodology applied to government-spending shocks reveals a mirror image: fiscal multipliers are *larger* when household debt is *low*. On impact, the multiplier is 0.70, 0.61, and 0.79 percentage points of GDP larger in Australia, Norway, and the United States, respectively, when the spending increase occurs during periods of low household indebtedness. The logic complements the monetary result: indebted households save rather than spend the extra income generated by fiscal expansion, dampening the Keynesian multiplier.[3]

[3] This result speaks directly to the timing of stimulus packages. The effectiveness of fiscal intervention is not constant—it depends on the household balance sheet.

Together, these two papers establish a single theme: *the household balance sheet is a first-order state variable for macroeconomic policy transmission*. Forecasts of how rate moves or spending packages feed through to aggregate demand should account for where in the household-debt cycle the economy sits, not just the size of the policy change.

How do indebted households adjust their labour supply?

Papers: Zurita (2025, working paper).

The macro evidence raised a micro question: *how* do indebted households absorb shocks at the individual level? Incomplete-markets models predict that agents self-insure through savings and labour supply, but the relative importance of these margins under heavy debt had received less attention.

Using Australian microdata, I document that middle-aged individuals whose annual mortgage repayments exceed 20 percent of household income significantly increase their labour supply in response to rising repayment obligations.[4] This finding—that debt pressure makes people work *more*, not less—has a macroeconomic implication. In an incomplete-markets model calibrated to the Australian economy, I study the transition dynamics following a productivity slowdown. Higher debt amplifies household reliance on labour income, generating a larger labour-supply response but also reducing the economy's resilience to further shocks. The household balance sheet not only affects demand-side transmission (as in my earlier papers) but also shapes the supply side of the economy.

[4] Australia provides an ideal laboratory: variable-rate mortgages dominate, so changes in the policy rate translate almost immediately into changes in household cash flows.

Does the distribution of bank equity matter for credit supply and growth?

Papers: Zurita (2026b, International Review of Economics and Finance).

The household-debt papers are about the *demand* side of credit. But credit has a supply side, and that supply is intermediated by banks. The question I turned to next was whether the structure of the banking sector—in particular, the distribution of bank equity—shapes how credit translates into economic activity.

Using newly compiled datasets covering more than 5,000 U.S. bank balance sheets alongside mortgage and business loan data, I show that the answer is emphatically yes.[5] Standard empirical models that treat all bank lending as homogeneous—ignoring who is lending—dramatically understate the economic effects of credit supply. Once bank equity-size heterogeneity is accounted for, the estimated impact of mortgage lending on local GDP growth can be up to *sixty times larger* than in a homogeneous-lender specification, while the effect of business lending can be up to eighteen times greater.

[5] Building this dataset was a substantial undertaking. U.S. Call Report data must be cleaned, merged with HMDA mortgage data, and matched to county-level economic outcomes.

The difference in amplification between mortgage and business lending is itself informative: credit allocation in the business-lending channel appears to be driven more by risk–return trade-offs than by bank size, whereas mortgage lending—more standardised and subject to GSE pricing—responds more mechanically to the balance-sheet capacity of the lender. This result connects to a broader theme in my agenda: the architecture of the financial system, not just its aggregate size, determines how credit affects the real economy.

Who bears the burden of bank capital regulation?

Papers: Zurita (2026c, job market paper).

If *which* banks lend matters for outcomes, then regulation that differentially affects banks will have heterogeneous real effects. My job market paper asks the distributional question directly: when regulators raise capital requirements, who pays?

Using narratively identified capital-requirement shocks and a quarterly panel of approximately 4,800 U.S. banks over twenty-five years (1994–2019), I document two cross-sectional facts. First, *within* banks, commercial lending contracts about forty percent more than real-estate lending following a tightening, reflecting the differential risk weights assigned to these asset classes under risk-based capital rules. Second, *across* banks, institutions with thin voluntary capital buffers—those operating closest to the regulatory minimum—absorb the largest share of the credit contraction.[6]

[6] The voluntary buffer is endogenous and reflects the bank's own precautionary motive. Banks with thin buffers are not necessarily the most fragile; they may simply face higher opportunity costs of holding idle capital.

To rationalise both facts jointly, I build a heterogeneous macro-banking model with precautionary buffers, solved globally and estimated on within-bank impulse responses. The model delivers a key quantitative result: a one-percentage-point increase in the capital requirement costs 2.4–2.9 percent of credit-market surplus. Crucially, redesigning risk weights or size tiers—the architecture of regulation—redistributes this burden across borrower types and bank sizes but barely changes its aggregate magnitude. The total cost is driven by the *level* of requirements, not their design.

This finding has a direct policy implication. Debates about capital regulation often focus on calibrating risk weights or carving out exemptions for community banks. My results suggest that these design choices matter for distribution—they determine *who* bears the burden—but the overall cost to credit markets is pinned down by how much capital regulators demand in total. The architecture is about fairness, not efficiency.

How does monetary policy transmit through bank profitability?

Papers: Zurita and Le (2026, submitted).

Regulation is not the only shock that hits banks asymmetrically. In joint work with Jiening Le, I study the profit channel of monetary policy: how do contractionary interest-rate surprises affect bank net interest income, and does the effect differ across the bank-size distribution?

Using high-frequency monetary-policy surprises matched to U.S. Call Report data, we identify a pronounced non-linearity. Contractionary shocks compress net interest income significantly more for medium-sized banks than for either large or small institutions. The mechanism is a *liability-substitution channel*: as policy rates rise, depositors shift from zero-cost current accounts to interest-bearing alternatives, forcing banks to replace cheap funding with expensive sources faster than the asset side reprices.[7] A one-standard-deviation increase in expense-channel exposure reduces net interest income by 9 percent for small banks and 11 percent for large banks, but the effect peaks at *19 percent* for medium-sized banks—institutions large enough to face competitive deposit markets but too small to access wholesale funding flexibly.

[7] This is related to, but distinct from, the "deposits channel" studied by Drechsler, Savov, and Schnabl (2017, *QJE*). Our focus is on the differential impact across the bank-size distribution, not just the aggregate deposit flow.

Local-projection estimates confirm that these profitability effects persist over several quarters, suggesting that monetary tightening generates lasting stress precisely in the middle of the bank-size distribution. Combined with the findings from my banking-heterogeneity paper, this result implies that a tightening cycle squeezes credit supply disproportionately through mid-sized lenders, with knock-on effects for the local economies they serve.

Can productivity growth substitute for fiscal discipline?

Papers: Zurita (2025b, working paper).

A separate strand of my work concerns fiscal sustainability in the United Kingdom. A common argument in policy circles is that faster productivity growth can "grow the economy out of" its fiscal challenges. This paper tests the claim.

In a dynamic stochastic general equilibrium (DSGE) framework, I study how productivity-driven growth interacts with fiscal-policy choices. The central result is negative: maintaining government spending per worker without increasing the debt-to-output ratio requires higher taxation to satisfy the intertemporal budget constraint. Productivity gains help, but they cannot replace fiscal discipline.

The tax-mix analysis delivers practical guidance. Capital-income tax increases, while effective in restoring fiscal balance, hinder capital accumulation and long-run growth—a cost that compounds over time. Consumption tax hikes generate substantial short-term volatility in the spending-to-output ratio. Labour tax increases create fewer distortions, because households can partially adjust by reallocating between work and leisure.[8] These findings contribute to the UK fiscal debate, but the framework applies to any advanced economy facing the tension between spending commitments, debt sustainability, and the political economy of taxation.

[8] The relative ranking of tax instruments is sensitive to the labour-supply elasticity. The household-debt paper above suggests that indebted households have limited ability to adjust labour supply, which would tilt the ranking further.

Where is the agenda heading?

My ongoing and planned research deepens the central theme—that the *distribution* of balance sheets determines how policy works—in two directions.

Earnings risk and business cycles. In ongoing work with Emiliano Carlevaro, I study how earnings risk varies over the business cycle in a volatile economy and how this interacts with household portfolio choices and credit demand. This project brings together the household-debt and banking strands of my agenda by closing the loop: if earnings risk is countercyclical, households demand more credit precisely when the banking sector is least willing to supply it.

Bridging household debt and bank regulation. So far, my work treats the household and bank sides of the credit market largely in isolation. The next step is a general-equilibrium model in which regulatory shocks to bank capital interact with the household debt cycle. The two amplification mechanisms should reinforce each other: tighter capital rules reduce credit supply, while high household debt reduces the demand-side absorption of fiscal attempts to offset the contraction.

What is the takeaway?

Several conclusions connect these projects:

1. **The household balance sheet is a state variable for policy transmission.** Monetary easing delivers a bigger short-run boost

when household debt is high, but the stimulus fades faster. Fiscal multipliers are larger when debt is low. Forecasting policy effects without conditioning on the debt cycle is incomplete.

2. **Bank heterogeneity amplifies credit-supply effects by an order of magnitude.** Ignoring which banks are lending leads to dramatic underestimates of the local economic impact of credit. The distribution of bank equity, not just aggregate lending, matters for growth.

3. **Capital regulation is about distribution, not just levels.** Risk weights and size tiers determine *who* bears the burden of higher capital requirements—commercial vs. real-estate borrowers, buffer-constrained vs. well-capitalised banks—but the aggregate welfare cost is pinned down by the overall stringency of the rule.

4. **Monetary tightening hits mid-sized banks hardest.** The liability-substitution channel generates a non-linear profitability squeeze concentrated in the middle of the bank-size distribution, with persistent effects on local credit supply.

5. **Indebted households work more, not less.** Debt pressure raises labour supply, making the economy more responsive to shocks on the extensive margin but also more fragile when further shocks arrive.

6. **Productivity growth cannot substitute for fiscal discipline.** The tax mix matters: labour taxes are the least distortive instrument for the UK's fiscal adjustment, while capital taxes create compounding long-run costs.

The thread connecting these results is that the questions most urgent in policy debates—who bears the cost of regulation? does debt make stimulus less effective?—are often where standard models are silent, and where careful empirical and quantitative work can make a difference.

Papers and Work in Progress

The table below lists the papers produced or in progress. Some are discussed in this essay; some are not.

Paper	Coauthors	Status
Household Debt and Macroeconomic Policy		
Does Household Debt Affect Monetary Policy Transmission?	—	Published, *Macro. Dynamics* 2026
Does Household Debt Affect the Fiscal Multiplier?	—	Published, *Macro. Dynamics* 2024
Household Debt and Low Labour Productivity	—	Working paper, 2025
Banking, Credit Supply, and Regulation		
Banking Heterogeneity, Credit Supply and Growth	—	Published, *IREF* 2026
Who Bears the Burden of Bank Capital Regulation?	—	Job market paper, 2026
The Bank Profit Channel of Monetary Policy	Le	Submitted, 2026
Fiscal Policy		
Labour Productivity and Fiscal Sustainability in the UK	—	Working paper, 2025
Ongoing Projects		
Earnings Risk and Business Cycles in a Volatile Economy	Carlevaro	In progress